Nex-Gen Lumina — Admin Operations Guide

This guide is for **you, the owner of Nex-Gen LED**. It covers how to set up and manage the entire dealer/installer ecosystem from within the Lumina app. Your dealers and installers have their own guide — this one is about the admin operations they can't see.

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1. Your Admin Credentials

Item	Value
Admin PIN	`9999`
Master Installer PIN	`8817` (Nex-Gen Administrator bypass)
Dev/Test PIN	`0000` (development only — disable before launch)

> **Before launch:** The admin PIN is currently hardcoded. Plan to move this to a Firebase Auth role check or a secure remote config value so it can be changed without an app update.

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2. Accessing the Admin Portal

1. Open the Lumina app on any device
2. Go to **Installer Mode** (from the login screen or Settings)
3. Tap **Admin Access**
4. Enter your admin PIN: `9999`
5. You land on the **Admin Dashboard**

The dashboard shows three live stats at the top:

- **Active Dealers** — companies you've onboarded
- **Active Installers** — technicians across all dealers
- **Installations** — total completed customer setups

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3. The Big Picture: How It All Fits Together

```
YOU (Admin PIN 9999)
│
├── Dealer 01: "ABC Lighting Co."
│    ├── Installer 01 → PIN 0101 (Mike)
│    ├── Installer 02 → PIN 0102 (Sarah)
│    └── Installer 03 → PIN 0103 (Jake)
│
├── Dealer 02: "Premier Outdoor LLC"
│    ├── Installer 01 → PIN 0201 (Tony)
│    └── Installer 02 → PIN 0202 (Lisa)
│
└── Dealer 03: "Bright Ideas Inc."
     └── Installer 01 → PIN 0301 (Carlos)
```

- **You** create **Dealers** (companies that sell/install your product)
- Each **Dealer** gets a 2-digit code (01–99) → up to 99 dealers
- Under each Dealer you create **Installers** (their technicians)
- Each **Installer** gets a 2-digit code (01–99) → up to 99 per dealer
- The installer's **4-digit PIN** = dealer code + installer code
- Installers use that PIN to enter Installer Mode and complete customer setups

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4. Creating Your First Dealer

Step-by-Step

1. From the Admin Dashboard, tap **Manage Dealers**
2. Tap the **+ Add Dealer** button
3. Fill in the dealer's information:

Field	Required	Notes
Contact Name	Yes	Primary contact at the company
Company Name	Yes	Their business name
Email	Yes	For correspondence
Phone	Yes	Business phone

4. The system **auto-assigns the next available dealer code** (starting at `01`)
5. Tap **Save**

The dealer is now active and visible in the dealer list.

> **Tip:** Dealer codes are assigned sequentially. If you've used 01, 02, 03 and deactivated 02, the next new dealer still gets 04 (not 02). Codes don't recycle — this prevents PIN collisions with old installers.

What to Tell the Dealer

After creating their account, give them:

- Their **dealer code** (e.g., `03`)
- The **Dealer & Installer Setup Guide** PDF (you already have this at `docs/Dealer_Installer_Setup_Guide.pdf`)
- Explain that you'll create their initial installer PINs, or give them admin access to do it themselves

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5. Adding Installers Under a Dealer

Step-by-Step

1. From the Admin Dashboard, tap **Manage Installers**
2. If you have multiple dealers, select which dealer this installer belongs to
3. Tap **+ Add Installer**

4. Fill in the installer's information:

Field	Required	Notes
Full Name	Yes	Installer's name
Email	Yes	Their email
Phone	Yes	Their phone

5. The system auto-assigns the next installer code under that dealer
6. Tap **Save**
7. The **4-digit PIN** is displayed — this is what the installer uses in the field

Example

You're adding the first installer under Dealer 03 ("Bright Ideas Inc."):

- System assigns installer code: `01`
- Combined PIN: `0301`
- You tell Carlos: "Your PIN is 0301"

Carlos opens the Lumina app → Installer Mode → enters `0301` → he's in.

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6. Day-to-Day Management

Viewing All Dealers

Admin Dashboard → **Manage Dealers** shows every dealer with:

- Dealer code
- Company name
- Contact name
- Active/inactive status
- Number of installers under them

Viewing All Installers

Admin Dashboard → **Manage Installers** shows every installer with:

- Their 4-digit PIN
- Name
- Parent dealer
- Active/inactive status
- Total installations completed

You can filter by dealer to see only one company's team.

Editing a Dealer or Installer

Tap any dealer or installer in the list to edit their contact details (name, email, phone). The dealer code and installer code **cannot be changed** after creation.

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7. Deactivating & Reactivating

Deactivating an Installer

Tap the installer → toggle **Active** to off.

- Their PIN stops working **immediately**
- Any in-progress installations they had are preserved (can be resumed by another installer)
- Their completed installation records remain intact

Deactivating a Dealer

Tap the dealer → toggle **Active** to off.

> **Cascade effect:** Deactivating a dealer **automatically deactivates ALL installers** under that dealer. Every one of their PINs stops working immediately.

Reactivating

Toggle **Active** back to on. For dealers, you must **manually reactivate each installer** — the cascade only works one direction (deactivation).

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8. What Happens During an Installation

When an installer uses their PIN and completes a setup, the system:

1. Creates a **Firebase Auth account** for the customer (email + temp password)
2. Creates a **user profile** in Firestore at `/users/{uid}`
3. Creates an **installation record** at `/installations/{id}` containing:
    - Customer info (name, email, address)
    - Your dealer code and installer code
    - Installer name and dealer company name
    - Date installed
    - Warranty expiration (5 years from install date)
    - Controller serial numbers
    - Site mode (residential or commercial)
4. Links the physical controllers to the customer's account
5. Increments the installer's installation count
6. Shows the customer's login credentials on screen for the installer to hand off

You can view all installation records from the Firebase Console under the `installations` collection.

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9. Firebase Collections You Own

Collection	What's In It	Who Writes
`/dealers/{id}`	Dealer companies (code, name, contact, active status)	You (admin)
`/installers/{id}`	Installer accounts (PIN, name, dealer, active status)	You (admin)

`/installations/{id}`	Completed installation records (customer, warranty, controllers)	Installers (during setup)
`/users/{uid}`	Customer accounts (profile, preferences, linked controllers)	Installer creates; customer owns
`/users/{uid}/properties`	Customer properties (homes, locations) with linked controller IDs	Customer creates/manages
`/dealerDemoCodes/{id}`	Demo/trial access codes for media team	You (admin, via Firebase Console)

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10. Media Mode (Content Team Access)

Media Mode is separate from Installer Mode. It lets your content/marketing team access any customer's system for video shoots without giving them installer privileges.

How It Works

1. You create a **demo access code** in the `dealerDemoCodes` Firestore collection
2. The media person opens Lumina → **Media Mode** → enters the 6-character code
3. They can search for any customer by email or address
4. They can **control the lights** but **cannot modify settings or create accounts**

Creating a Media Access Code

Currently done directly in Firebase Console:

1. Go to Firestore → `dealerDemoCodes` collection
2. Add a document with these fields:

Field	Type	Example
`code`	string	`MEDIA1`

`dealerCode`	string	`00` (use 00 for internal Nex-Gen)
`dealerName`	string	`Nex-Gen LED`
`market`	string	`Internal`
`isActive`	boolean	`true`
`usageCount`	number	`0`
`maxUses`	number	`null` (unlimited) or a number
`createdAt`	timestamp	(current time)
`expiresAt`	timestamp	`null` (never) or a date

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11. Firestore Security Rules Summary

Your security rules enforce this access model:

Who	Can Do
Regular users	Read/write only their own data (profile, controllers, properties, schedules, geofences)
Media/dealer/admin users	Read any user's data including properties (for customer lookup and content)
Installers	Create installation records (with required fields)
Admins	Full CRUD on dealers, installers, installations

The `user_role` field on each user's profile (`residential` , `media` , `dealer` , `admin`) determines their access level.

Subcollections under `/users/{uid}` with owner-only write access and media/dealer/admin read access:

- `controllers` — registered WLED controllers
- `properties` — user's properties/locations with linked controller IDs
- `commands` — cloud relay commands (bridge can also read/write)
- `bridge_status` — ESP32 bridge heartbeat
- `geofences` — location-based automation triggers

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12. Onboarding a New Dealer — Complete Checklist

Here's the full process when you sign up a new dealer:

- ☐ **Get their info**: contact name, company name, email, phone
- ☐ **Open Admin Portal**: Lumina app → Installer Mode → Admin Access → PIN `9999`
- ☐ **Create the dealer**: Manage Dealers → Add Dealer → fill in info → Save
- ☐ **Note the dealer code** (e.g., `03`)
- ☐ **Create their installers**: Manage Installers → select dealer → Add Installer for each tech
- ☐ **Note each PIN** (e.g., `0301`, `0302`, etc.)
- ☐ **Send the dealer**:
  - Their dealer code
  - Each installer's 4-digit PIN
  - The **Dealer & Installer Setup Guide** PDF
- ☐ **Confirm they can log in**: Have one installer test their PIN in Installer Mode

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13. Troubleshooting

"Maximum dealer limit (99) reached"

You have 99 dealer codes used. You'd need to delete or reclaim unused ones, or expand the code format. This is a code change.

An installer says their PIN doesn't work

1. Check the Admin Portal — is their account set to **Active**?
2. Check their parent dealer — is the **dealer** active? (Deactivated dealer = all PINs dead)

3. Verify they're entering all 4 digits correctly
4. If they've been locked out (5 failed attempts), the lockout resets after restarting the app

A dealer left the program

1. Admin Portal → Manage Dealers → find them → toggle **Active** off
2. All their installers are automatically deactivated
3. Existing customer installations are unaffected — those accounts still work
4. The dealer code is not reused

Need to transfer an installer to a different dealer

You can't move an installer between dealers — the dealer code is baked into their PIN. Instead:

1. Deactivate the old installer account
2. Create a new installer under the new dealer
3. Give the technician their new PIN

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14. Security Notes

Current limitations to address before scaling:

- **Admin PIN is hardcoded** (`9999` in `admin_providers.dart`). Anyone with the app who guesses it has full admin access. Move to Firebase Auth custom claims or remote config.
- **Master installer PIN is hardcoded** (`8817` in `installer_providers.dart`). Same concern.
- **Demo/media codes** are managed in Firebase Console, not in the app. Consider adding an in-app screen for this.
- **No audit log** — dealer/installer changes are not tracked. Consider adding a Firestore `audit_log` collection.
- **Deletes are soft deletes** — deactivation only. Data is preserved.

15. Quick Reference

Action	Where
Enter Admin Portal	Installer Mode → Admin Access → PIN `9999`
Create a dealer	Admin Dashboard → Manage Dealers → Add
Create an installer	Admin Dashboard → Manage Installers → Add
Deactivate a dealer	Manage Dealers → tap dealer → toggle Active off
Deactivate an installer	Manage Installers → tap installer → toggle Active off
View installation stats	Admin Dashboard (top cards)
Create media access code	Firebase Console → `dealerDemoCodes` collection
View all installations	Firebase Console → `installations` collection

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